

CSE 123A - Week 6 Status

Group 8

Link to Status Updates section in our repo

https://git.ucsc.edu/pnauwela/123a-project/-/tree/main/Status%20Updates?ref_type=heads

What we Did

What we are Working On

What's next

Individual Contributions

Dev Chodavadiya:

This week I focused on hardware iteration and documentation.

On the hardware side, I reprinted the base plates to fix a flatness issue — the previous prints were slightly slanted, so I adjusted the print settings to ensure the bottom plate sits level on a surface.

On the documentation side, I made progress on the design document by writing the research into existing designs section, the sustainability statement, and the aesthetic prototype section with CAD renders and photos of the current assembly.

I will design and print washers to go between the load cell sensor and the plates for better contact and stability. The team will continue working on the demo script to have it ready for the demonstration.

Logan Bossuwe:

Pieter Nauwelaerts:

This week I created the testing code for the HX711 sensor. The major changes include a different way of approaching how we design the microcontroller code in order to allow us to switch between the microcontroller, and test code. This was done using a C header file to define the shared functions, and having two separate main files, one for microcontroller deployment, and one for testing.

The test code simulates the Wheatstone bridge sensor and passes the functions data that would come in or out. Simulated weight readings go into the functions, the logic is the same as the logic used by the microcontroller, and the output comes out as is verified against what

we should expect to see. The logic we use in the microcontroller can now be verified and tested. This includes pushing it to the limit by having very fast sequences of activity and checking that the outputs come out in the same order.

Reid Graham:

This week, I created the code for the microcontroller to connect to a WiFi network and send HTTP GET and POST requests. I also created header files for organization and relevant functions, as well as helped to integrate this functionality into the main file. I also created some test code for the WiFi provisioning code, including testing important functions like url decoding. For the physical sensor, I installed a new base plate for the scale that made it more stable.

I also worked on the design document, including editing the architecture diagram, adding to the block diagrams section, and writing some of the documentation for the WiFi provisioning functionality.

For the next week, I want to prioritize finalizing the design document, including the sections on test plan and results and MCU HTTP functionality.

Sam Lai:

Shriyan Gote: